Credit Card Fraud Detection: Exploratory Data Analysis and Behavioral Profiling

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***Abstract*—Credit card fraud detection is a critical problem in modern financial systems due to the increasing volume of digital transactions and evolving fraud strategies. This study presents a comprehensive exploratory data analysis (EDA) of transactional data to identify behavioral, temporal, and categorical patterns associated with fraudulent activity. The analysis reveals that fraud is strongly associated with late-night transactions, higher transaction amounts, and specific transaction categories such as online shopping. Additionally, weak linear correlations between numeric features and fraud indicate that fraud detection is driven by complex interactions rather than individual predictors. These findings provide a strong foundation for subsequent machine learning modeling.**

***Index Terms*—Fraud Detection, Exploratory Data Analysis, Credit Card Fraud, Data Science, Behavioral Analysis**

I. Introduction

The rapid adoption of digital payment systems has significantly increased both transaction convenience and exposure to fraud. Credit card fraud detection has therefore become a major concern for financial institutions.

Traditional rule-based detection systems struggle to adapt to evolving fraud patterns. As a result, data-driven approaches have become essential. However, before applying machine learning techniques, it is critical to understand the structure and behavior of the data.

Exploratory Data Analysis (EDA) plays a key role in:

- uncovering hidden patterns,
- identifying high-risk behaviors,
- understanding feature relationships,
- guiding feature engineering.

This study focuses exclusively on understanding fraud patterns through EDA, forming a foundation for future modeling work.

II. Methodology

The analysis follows a structured exploratory approach:

1) Data preprocessing and cleaning
2) Feature engineering (time-based and demographic features)
3) Fraud rate analysis across key dimensions
4) Fraud-only behavioral profiling
5) Correlation analysis of numeric features

Key engineered features include:

- **Age bands** for demographic grouping
- **Transaction hour** and **day of week**
- **Night transaction indicator**

The target variable represents whether a transaction is fraudulent.

III. Results

A. Temporal Fraud Patterns

	total_transactions	fraud_transactions	fraud_rate_pct
Daytime	1297347	1341	0.1034
Late Night	555047	8310	1.4972

Fig. 1: Fraud Rate: Daytime vs Late Night

Fraud rates are significantly higher during late-night periods compared to daytime transactions, suggesting that fraudulent activity is more prevalent during low-surveillance hours.

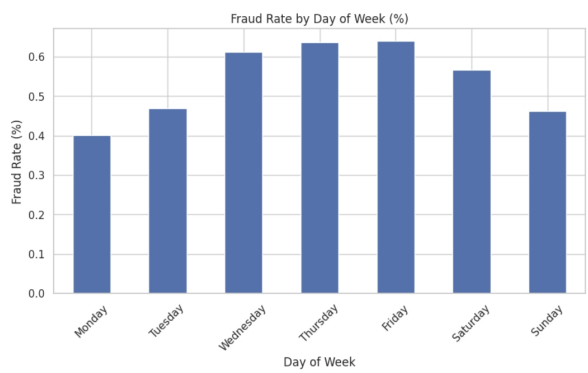

Fig. 2: Fraud Rate by Day of Week

Fraud rates increase gradually throughout the week, peaking on Thursday and Friday. This may reflect increased transaction activity and spending behavior toward the end of the week.

B. Demographic Analysis

The dataset is dominated by middle-aged individuals, with most customers between 30 and 60 years old.

There is substantial overlap in age distributions between fraudulent and non-fraudulent transactions, indicating that age alone is not a strong predictor of fraud.

Fraud risk varies across age groups, with higher rates observed in older customers and moderate risk among younger users.

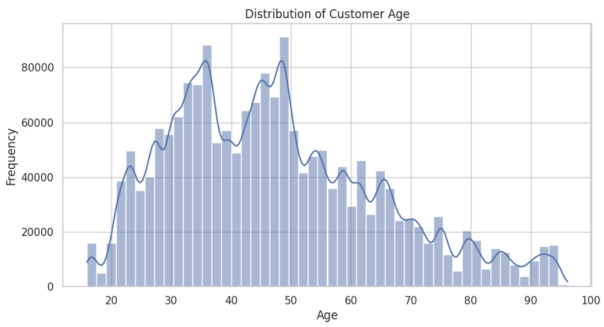

Fig. 3: Customer Age Distribution

Fig. 4: Customer Age by Fraud Class

C. Transaction Characteristics

Fraudulent transactions tend to occur at higher monetary values, suggesting that fraudsters target larger transactions to maximize gains.

Fraud is concentrated in specific transaction categories, particularly online shopping and point-of-sale transactions.

D. Correlation Analysis

The correlation analysis shows that most numeric features have weak linear relationships with fraud, indicating that fraud detection is not driven by single-variable effects.

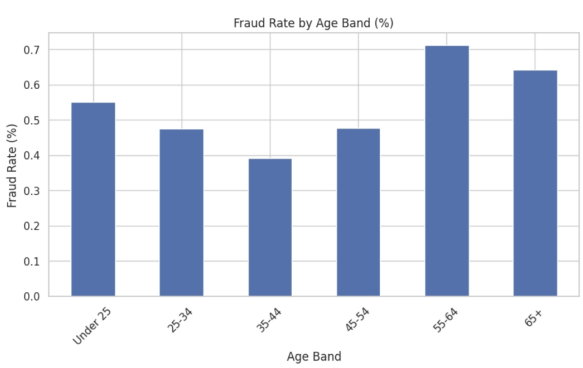

Fig. 5: Fraud Rate by Age Band

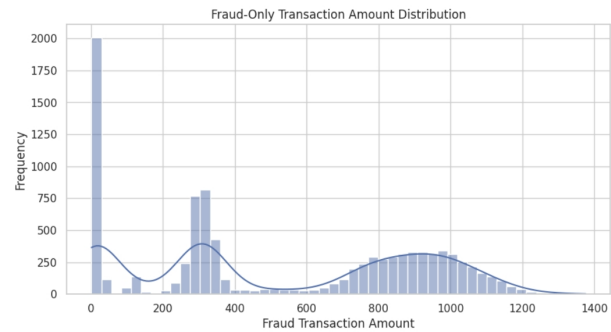

Fig. 6: Fraudulent Transaction Amount Distribution

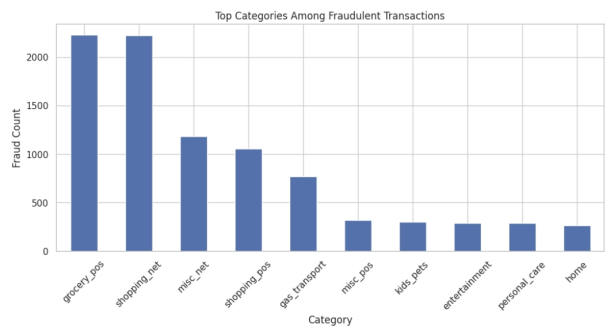

Fig. 7: Top Categories Among Fraudulent Transactions

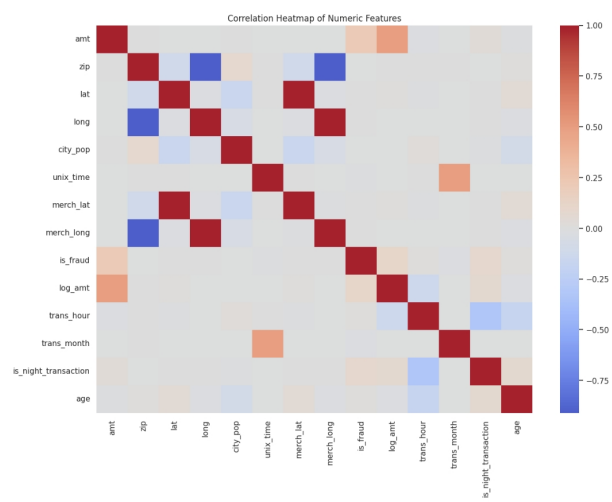

Fig. 8: Correlation Heatmap of Numeric Features

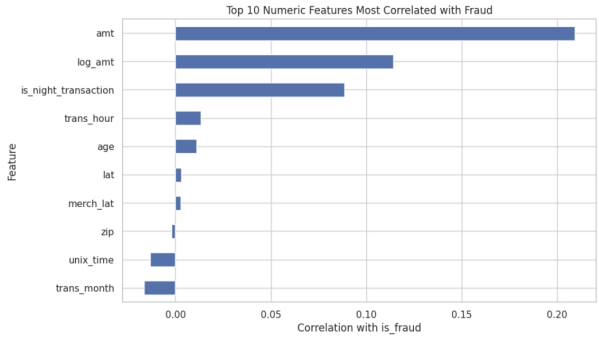

Fig. 9: Top Numeric Features Correlated with Fraud

Transaction amount and night transaction indicators emerge as the most relevant features, although their correlations remain moderate.

IV. Discussion

The results highlight that fraudulent behavior is primarily driven by transaction context rather than customer demographics.

Key insights include:

- Fraud is significantly higher during late-night hours.
- Fraud increases toward the end of the week.
- High-value transactions are more likely to be fraudulent.
- Fraud is concentrated in specific transaction categories.

These findings suggest that fraud detection requires combining multiple behavioral signals rather than relying on individual features.

V. Limitations

The analysis has several limitations:

- **Class imbalance:** Fraud cases are rare relative to legitimate transactions.
- **Limited feature scope:** Missing features such as device data, transaction velocity, and user history.
- **Static dataset:** Does not capture evolving fraud patterns over time.
- **Correlation limitations:** Only linear relationships are examined.

VI. Implications for Modeling

The findings from this EDA provide clear direction for future modeling:

- Use non-linear models such as Random Forest or XGBoost.
- Prioritize features related to transaction amount and timing.
- Engineer behavioral features such as night transaction indicators.
- Address class imbalance using techniques such as resampling or class weighting.

VII. Conclusion

This study demonstrates that credit card fraud is fundamentally a behavioral phenomenon rather than a purely demographic one. Fraudulent transactions are more likely to occur during late-night periods, involve higher transaction amounts, and appear in high-risk categories.

The weak individual correlations observed highlight the importance of capturing complex feature interactions. These insights provide a strong foundation for the development of robust machine learning-based fraud detection systems.

References

[1] Bolton, R. J., and Hand, D. J., "Statistical Fraud Detection: A Review," *Statistical Science*, 2002.
[2] Ngai, E. W. T., Hu, Y., Wong, Y. H., Chen, Y., and Sun, X., "The Application of Data Mining Techniques in Financial Fraud Detection," *Decision Support Systems*, 2011.
[3] Phua, C., Lee, V., Smith, K., and Gayler, R., "A Comprehensive Survey of Data Mining-Based Fraud Detection Research," 2010.
[4] Breiman, L., "Random Forests," *Machine Learning*, 2001.
[5] Kaggle, "Credit Card Fraud Detection Dataset."